

# AFREEN ALAM

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## EDUCATION

**Boston University**  
M.S. Applied Business Analytics

Boston, MA  
Dec '26

**National University of Singapore**  
B.Sc. Data Science and Analytics

Singapore  
May '22

## EXPERIENCE (3+ years)

### Massachusetts General Hospital (Home Base)

Boston, MA

#### **Data Analyst Co-op**

Jun '26 – Present

- Build and optimize Tableau dashboards, conduct ad hoc analytics and ETL processes, and improve data reliability through validation and root-cause analysis of data quality issues

### Boston University

Boston, MA

#### **Research Assistant (ShieldAI - Explainability tools taxonomy for AI risk mitigation)**

Jan '26 – Present

- Leading a collaboration with 3 professors to evaluate 20+ explainable-AI and risk-mitigation tools against enterprise AI risks, combining literature review and hands-on testing to drive a scientific manuscript on AI risk management

### Standard Chartered Bank

Singapore

#### **Data Analyst | (HF Transformers, LangChain, LLM, RAG, Python, Tableau)**

Oct '23 – Apr '25

- Onboarded and mentored an intern—accelerated ramp-up, connected her with team leads, coached project and executive presentation which ranked in the **top 5%** and resulted in a **full-time offer**
- Prototyped a RAG system (LangChain text-splitter benchmarking, DeepEval) using **100+** bank annual reports using **FAISS + Qwen2.5-1.5b**; improved top-5 retrieval precision by **35%** (52→87%), and cut analyst review time by **45%**
- Built a Python synthetic-data pipeline (Faker) from **10k** cross-customer sample to generate **50k+** privacy-compliant records, preventing data leaks, avoiding demo downtime, and preserving trust of **950+** top-tier client accounts
- Built an interactive Tableau dashboard that segmented **12K+** loan accounts by utilization behavior, surfaced trends and alerts for relationship managers, and enabled targeted limit rebalances that cut underutilized exposure by **20%**, improving portfolio utilization and client engagement

#### **Digital Channels and Data Analytics Graduate | (R, NLTK, Google Analytics)**

Sept '22 – Oct '23

- Prototyped an XGBoost cash-balance forecasting model (MAPE **14%**) targeted for integration into a treasurer-facing analytics dashboard; enabling ~\$30M/year in working-capital optimization per platinum client by reducing idle cash buffer by 43%, and accelerating onboarding of platinum clients (> \$100M revenue)
- Analyzed Google Analytics event-level data using funnel analysis to quantify user drop-off across digital banking journeys, informing UX changes that improved user journey completion by **48%**
- Engineered an NLP solution (n-grams, frequency analysis) on **10K+** mobile-banking feedback records to surface pain points driving **86%** of negative sentiment; prioritized fixes that reduced the negative feedback to **14%**

### Biofourmis

Singapore

#### **Data Science Intern | (Statistics, PyTorch, Python)**

Dec '21 – May '22

- Led a Bland–Altman analysis of physiological time-series from **100k+** patients, comparing wearable sensors to clinical reference signals; **95%** of measurements fell within limits of agreement, validating clinical-grade consistency
- Engineered and adapted a PyTorch CNN (IoU **0.78**) trained on 2-second ECG chunks from **105** signals across **290** patients to predict waveform indices for **irregular arrhythmia** detection, enabling early anomaly alerts and real-time pilot deployments

## RESEARCH PROJECTS / HACKATHON

### RAG System for SEC 10-K | (FAISS, LangChain, RAGAS, OpenAI API, BeautifulSoup, Regex)

Jan '26 – May '26

- Built a section-scoped Retrieval-Augmented Generation (RAG) pipeline for SEC 10-K filings, leveraging NLP, transformer embeddings, and semantic search to extract AI strategy and risk insights from 7,700+ financial reports across 725 companies

### Smart Home IoT Data Science Project | (Python, SQLite)

Sep '25 – Dec '25

- Built an ML pipeline with SQL-based feature engineering for device-failure prediction; optimized classifiers via stratified CV and hyperparameter tuning (best AUC-ROC **0.82**), surfacing insights in Plotly dashboard

## TECHNICAL SKILLS

- **LLM/ GenAI:** LangChain, Hugging Face Transformers, Reranker, OpenAI API, GitHub Models API, FAISS, RAGAS
- **Machine Learning Frameworks:** PyTorch, Scikit-learn
- **Visualization:** Tableau, Power BI, Plotly, Matplotlib
- **Programming Languages:** Python, SQL, R